MILITARY SPECIFICATION

CIRCUIT BREAKERS, MANUAL AND
AUTOMATIC (28 VOLTS DC)

This amendment forms a part of Military Specification MIL-C-13516E, dated 28 March 1979, and is approved for use within the U.S. Army Tank-Automotive Command, Departments and Agencies of the Department of Defense.

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2.1, under "SPECIFICATIONS, MILITARY", delete:

"MIL-E-13856 - Electrical Components for Automotive Vehicles, Waterproofness Tests."

2.1, under "SPECIFICATIONS, MILITARY", add:

"MIL-I-81023 - Inductor, 28 V.D.C. Laboratory Test, General Specification for."

2.1, under "STANDARDS, MILITARY", add:

"MIL-STD-1184 - Electrical Components for Automotive Vehicles, Waterproofness Tests."

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Table I, first column, delete "65° ± 5°F" and substitute "-65 ± 5°F".

Table I, third column, delete "185" and substitute "196".

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Table I, third column, delete "115" and substitute "122".

3.4.6, line 2, delete "130" and substitute "138".

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* 3.6, line 6, delete "Military part number" and substitute "PIN".

Add new paragraph:

"4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

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4.3, add to end of first sentence", tolerance on inductors shall be -10 percent, +50 percent, tolerance on currents shall be ±10 percent, and tolerance on resistors shall be ±10 percent."

Table II, delete the column heading entitled "Maximum accuracy tolerance" and substitute "Maximum tolerance".

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4.6.6.2, line 7, delete "130" and substitute "138".

4.6.7, line 3, delete "4.5.6" and substitute "4.6.5".

4.6.7, in the tabular section, delete the first column and substitute:

"Classification
Type I - Class 1
Type I - Class 2
Type II - Class 1
Type II - Class 2"

4.6.7, in the tabular section, delete the column heading entitled "Minute cycles" and substitute "Minimum cycles".

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Paragraph 4.6.9. delete in its entirety and substitute:

"4.6.9 Endurance test. To determine conformance to 3.4.9, the circuit breaker shall be connected as herein specified and calibrated as specified in 4.6.5 at 200 percent of rated current. Type I circuit breaker shall operate for 500 cycles (make and break) and type II for 2000 cycles, and shall subsequently be subjected to 105 percent and 145 percent of rated current. During testing, circuit breakers shall be observed to determine conformance to 3.3.3.1. One of the following alternate test circuits (paragraph 4.6.9.1 or 4.6.9.2) shall be used.

4.6.9.1 Alternate A. The circuit breaker shall be connected as shown in figure 2a. The values selected for R_1, R_2, and L_1 shall be such that:

$$\frac{L_1}{R_1 + R_L} = 0.026 \pm 0.004$$

$$R_1 + R_L = R_2$$

Where:

R_1 = Variable resistance in right leg of circuit.

R_L = Resistance of coil L_1 in right leg of circuit.

R_2 = Variable resistance in left leg of circuit.

L_1 = Inductance of coil L_1 in right leg of circuit.

0.026 = Dimensionless time constant.

4.6.9.2 Alternate B. The circuit breaker shall be connected as shown in figure 2b. R_1 shall be adjusted to give the specified current. L_1 shall be an inductor conforming to type I, MIL-I-81023. When it is necessary to parallel inductors to achieve specified circuit breaker tests currents, paralleling shall be in accordance with MIL-I-81023."

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4.6.14. line 3. delete "MIL-E-13856" and substitute "MIL-STD-1184".

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Add new paragraph:

"6.7 Subject term (key word) listing.

Automatic circuit breakers
Circuit breakers
Manual circuit breakers"

Concluding material. after "Review activities". delete "ER" and "MU", and add "AR".

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* Add new Paragraph 6.8:

"6.8 Part identification number (PIN). The PINs to be used for circuit breakers acquired to this specification are constructed as follows:

M 13516/ X X X - XXX

- XXX — Amperage rating
- X — Type I - Manual reset; II - Automatic reset
- X — Grade A - Waterproof; B - Nonwaterproof
- X — Specification sheet no.
- 13516/ — Military specification no.
- M — Inch-pound units specification"

Renumber "Figure 2" as "Figure 2a".

Add new figure:

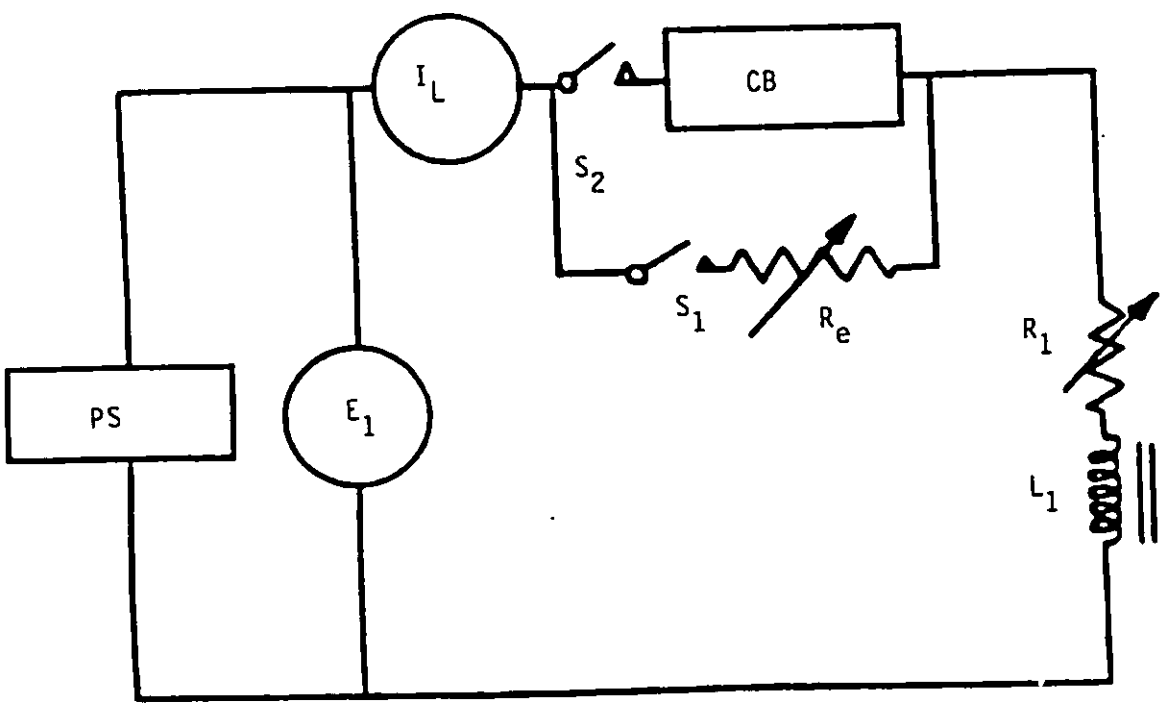


FIGURE 2b. Endurance test.

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Custodians:
Army - AT
Navy - MC
Air Force - 85

Review activities:
Army - AR, CR, ER
Air Force - 99
DLA - ES

User activities:
Army - AV, ME

Preparing activity:
Army - AT

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